higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

BUILDING AND STRUCTURAL SURVEYING N5

26 JULY 2019

This marking guideline consists of 4 pages.

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SECTION A

QUESTION 1

1.1 1.1.1 Lowest level✓ in a pipeline✓ on which water will flow✓

1.1.2 Distance✓ between two✓ adjacent contour lines✓

1.1.3 Common site name✓ given to any instrument✓ that can be mounted on a tripod✓

1.1.4 Position of a point✓ from a point of zero origin✓✓

1.1.5 Process of measuring✓ distance horizontally✓ on the surface of the earth✓

(5 × 3) (15)

1.2 1.2.1 True
1.2.2 False
1.2.3 True
1.2.4 True
1.2.5 True

(5 × 2) (10)
[25]

QUESTION 2

2.1 C1–C2 = ✓ $[(-75263.37 - -75243.16)^2$✓ + $(3649\,934.32 - 3650\,011.00)^2]$✓
= ✓ $[(-20.21)^2 + (-76.68)^2]$ ✓
= 79.300 m✓✓

C1– C3 = ✓ $[(-75316.13 - -75\,292.89)^2$✓ + $(3650\,016.79 - 3650\,010.66)^2]$✓
= ✓ $[(-23.24)^2 + (6.13)^2]$ ✓
= 24.033 m✓✓

C3–C4 = ✓ $[(-75\,309.79 - -75\,292.89)^2$✓ + $(3\,649\,946.55 - 3\,650\,010.66)^2]$✓

= ✓ $[(-16.900)^2 + (-64.11)^2]$ ✓
= 66.300 m✓✓

C3 – C7 = ✓$[(-75\,316.13 - -75\,292.89)^2$✓ + $(3650\,016.79 - 3650\,010.66)^2]$ ✓
= ✓$[(-23.24)^2 + (6.13)^2]$ ✓
= 24.033 m✓✓

C7 – C6 = ✓$[(-75\,333.84 - -75\,316.13)^2$✓ + $(3\,649\,966.33 - 3\,650\,016.79)^2]$✓
= ✓$(-17.71)^2 + (-50.46)^2]$ ✓
= 53.477 m✓✓ (25)

2.2
- Use the correct zero mark.
- Measure to the centre of a peg or a ranging rod.
- View the tape vertically on the graduations.
- Clear obstacles on the chain line.
- Minimise sag. (10)

2.3
- pegs
- ranging rod
- tape
- plumbing
- hammer
- lime

(Any 5 × 1) (5)
[40]

QUESTION 3

3.1

BACK SIGHT	INTER. SIGHT	FORE-SIGHT	RISE	FALL	REDUCED LEVEL	REMARKS
1.17					30.32	Benchmark 30.23
	1.42			0.25✓	29.98✓	A
2.18		1.76		0.34	29.64✓	B
	2.28			0.1	29.54✓	C
0.81		0.93		1.35✓	28.19✓	D
1.43		3.00		2.184✓	26.006✓	E
1.46		1.35	0.08✓		26.086✓	F
	1.35		0.11		26.196✓	G
	1.40			0.05✓	26.146✓	H
		0.52	0.88✓		27.026✓	I
1.07 4.274 -3.204		4.274			30.23 27.026 3.204	

3.204 – 3.204 = 0 there is no error (15)

3.2 At two points A and B erect perpendicular line BC,✓ at line BC erect another perpendicular line CD✓ to clear the obstacle. At CD erect a perpendicular line DE equal in length✓ to BC and at DE set off a right angle EF.✓ The direction EF is the extension of the survey line and distance CD = BE.✓ (5)
[20]

QUESTION 4

A –B 100.261 m at a slope of 3° 44'20"
Slope = 3,739°✓
AB = 100,261 × Cos 3,739°✓
= 100.048 m✓

B – C 72,408 m a slope of 7° 45'✓

Slope = 7,75°✓

BC = 72,408 m × Cos 7,75°✓

= 71.747 m✓

C –D = 50 m at a slope of 1° 51'✓

Slope = 1.85✓

CD = 50 × Cos 1.85✓

= 49,98 m✓

AD = 100,048 m✓ + 71,747 m✓ + 49,98 m✓

= 221,775 m✓ **[15]**

TOTAL: 100

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